$$\Delta G^{\circ} = - RT \ln K_{eq} \quad .....(9.26)$$

Comparing (9.25) and (9.26),

$$nFE^{\circ}_{cell} = RT \ln K_{eq}$$

$$\Rightarrow E^{\circ}_{cell} = \frac{2.303\ RT}{nF} \log K_{eq} \quad .....(9.27)$$

9.4.1 Nernst equation

Nernst equation is the one which relates the cell potential and the concentration of the species involved in an electrochemical reaction. Let us consider an electrochemical cell for which the overall redox reaction is,

$xA + yB \rightleftharpoons lC + mD$

The reaction quotient Q for the above reaction is given below

$$Q = \frac{[C]^{l}\ [D]^{m}}{[A]^{x}\ [B]^{y}} \quad .....(9.28)$$

We have already learnt that,

$$\Delta G = \Delta G^{\circ} + RT \ln Q \quad .....(9.29)$$

The Gibbs free energy can be related to the cell emf as follows [∴ equation (9.24) and (9.25)]

$$\Delta G = - nFE_{cell} \quad ; \Delta G^{\circ} = - nFE^{\circ}_{cell}$$

Substitute these values and Q from (9.28) in the equation (9.29)

$$(9.29) \Rightarrow - nFE_{cell} = - nFE^{\circ}_{cell} + RT \ln \frac{[C]^{l}[D]^{m}}{[A]^{x}\ [B]^{y}} \quad .....(9.30)$$

Divide the whole equation (9.30) by (-nF)

$$(9.25) \Rightarrow E_{cell} = E^{\circ}_{cell} - \frac{RT}{nF} \ln \frac{[C]^{l}[D]^{m}}{[A]^{x}\ [B]^{y}}$$

$$(or)\ E_{cell} = E^{\circ}_{cell} - \frac{2.303RT}{nF} \log \frac{[C]^{l}[D]^{m}}{[A]^{x}\ [B]^{y}} \quad .....(9.31)$$

The above equation (9.31) is called the Nernst equation

At 25°C (298K), the above equation (9.31) becomes,

$$E_{cell} = E^{\circ}_{cell} - \frac{2.303 \times 8.314 \times 298}{n(96500)} \log \frac{[C]^{l}[D]^{m}}{[A]^{x}\ [B]^{y}}$$

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n} \log \frac{[C]^{l}[D]^{m}}{[A]^{x}\ [B]^{y}} \quad .....(9.32)$$

$$\left[\begin{array}{l} \therefore\ R = 8.314\ JK^{-1}\ mol^{-1} \\ T = 298\ K. \\ 1\ F = 96500\ C\ mol^{-1} \end{array}\right]$$

Let us calculate the emf of the following cell at 25°C using Nernst equation.

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